MatLab Motif Tracking Workbook (Expanded Edition)

This workbook gives you five different ways to visualize movement motifs using MatLab. Each example shows a different perspective on motif patterns. Try running each script, then replace the sample data with your own motif codes.

Example 1: Histogram (Motif Frequency)

```
% Histogram of motif frequencies
motifs = [2 5 2 3 5 2 5 3 2];
histogram(motifs);
title('Motif Frequency');
xlabel('Motif Number');
ylabel('Count');
```

Example 2: Line Plot (Motif Sequence Over Time)

```
% Line plot of motif sequence over time
motifs = [1 2 1 3 2 4 1 3 2 2 4 1];
plot(motifs, '-o');   % '-o' = line with circles at each point
title('Motif Sequence Over Time');
xlabel('Time Step');   % when the motif occurred
ylabel('Motif Code');  % which motif was performed
```

Example 3: Bar Chart (Motif Frequencies)

```
% Bar chart of motif frequencies
motifs = [2 2 3 1 2 4 3 1 3 2 4 1 1];
counts = histcounts(motifs, 1:5);
bar(1:4, counts);
title('Motif Frequencies');
xlabel('Motif Code');
ylabel('Count');
```

Example 4: Pie Chart (Motif Distribution)

```
% Pie chart of motif distribution
motifs = [1 1 1 2 2 3 3 3 3 4 4];
counts = histcounts(motifs, 1:5);
pie(counts, {'Motif 1', 'Motif 2', 'Motif 3', 'Motif 4'});
title('Motif Distribution');
```

Example 5: Reading Motifs from a CSV File

```
% Example: Reading data from a CSV file

% If your CSV has one column of motif codes (no header):
motifs = readmatrix('motifs.csv');

% If your CSV has a header called 'Motif':
T = readtable('motifs.csv');
motifs = T.Motif;

% Plot histogram from CSV data
histogram(motifs);
title('Motif Frequency from CSV');
```

```
xlabel('Motif Number');
ylabel('Count');
```

Your Turn!

1. Create a CSV file of your own motif codes (one number per line). 2. Load your CSV into MatLab using Example 5. 3. Generate all five visualizations (histogram, line plot, bar chart, pie chart, CSV-driven histogram). 4. Save each plot as an image (.png or .jpg) and upload it to your GitHub archive.